SHAIRA AFRA REDHILA

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RESEARCH INTERESTS

Biomaterials and Bio-nanomaterials; Hydrogel-based Drug Delivery and Wound Healing; Tissue Engineering and Regenerative Medicine; Biomedical Nanotechnology; Biosensors and Biomedical Devices; AI/ML Applications in Healthcare.

EDUCATION

Khulna University of Engineering & Technology *CGPA: 3.84/4.00 (Ranked 4th in overall merit position)*

Bachelor of Science in Biomedical Engineering *March, 2022- September, 2026*

- **UG THESIS: "**Development and Optimization of Physically Cross-linked Chitosan/PVA/Gelatin Composite Hydrogels for Antibiotic Loaded Wound Dressing Applications"
- Systematically evaluated the effects of polymer stock concentration and Chitosan/PVA/Gelatin ratio on hydrogel properties for wound-dressing applications.
- Investigated swelling, water retention, gel fraction, pH responsiveness, mechanical properties, WVTR, porosity, biodegradation, and structural characteristics to establish formulation–property relationships.
- Selected promising formulations for SEM/XRD, BSLA, antibacterial testing, and ciprofloxacin loading/release toward application-specific wound-care optimization.

Higher Secondary Certificate (HSC) | GPA: 5:00 *2018- 2020*

Viqarunnisa Noon School & College (English Version)

Secondary School Certificate (SSC) | GPA: 5:00 *2008-2018*

Mohammadpur Preparatory School & College (English Version)

PUBLICATIONS

Conference Publications

- S. B. Shuvo, S. Hossain, S. A. Redhila, and U. Bose, "Exploring Common Molecular Interactions across Multiple Cancers to Identify Potential Therapeutic Targets and Drug Candidates," EICT 2025, KUET.
- S. Datta, G. M. M. Rahman, P. Majumder, S. A. Redhila, et al., "A Voice-Assisted Smart Device for the Visually Impaired People: Leveraging AI Vision, Object Detection, and GPS-Based Navigation," IEEE BECITHCON 2025.

Journal Manuscripts

- Thesis-based manuscript (in preparation).

PROJECTS

AI-Driven Glucose Monitoring System for Telemedicine Application | *BME 4232, KUET* | May 2026

- Designed a low-cost spectrophotometric glucose as an alternative to costly lab equipment for resource-constrained telemedicine use

Temperature-Regulated CPAP System for Sleep Apnea Management | *BME 3200, KUET* | Feb-May 2024

- Engineered a thermistor-based CPAP prototype that auto-adjusts air pressure based on breathing-cycle temperature changes

Arterial Wall Mechanics | *BME 3114, KUET* | Feb-May 2024

- Built a COMSOL computational model of arterial walls using the Holzapfel-Gasser-Ogden (HGO) framework to simulate hyperelastic and viscoelastic behavior of multi-layer, fiber-reinforced tissue

Implementation of a Digital Hearing Aid System using MATLAB | *BME 3112, KUET* | May 2024

- Simulated a full digital hearing-aid signal chain: noise injection, wavelet vs. lowpass denoising, FFT-based frequency-shaping/gain modification, and amplitude normalization

Implementation of an Artificial Pacemaker | *BME 2232, KUET* | Oct-Nov 2023

- Designed and simulated a basic pacemaker circuit to detect abnormal ECG rhythms and deliver corrective electrical pulses, demonstrating feasibility of low-cost cardiac support systems

TRAINING

- Training on Laboratory Animals Used in Biomedical Sciences | icddr,b, Aug 2026
- Industrial Attachment in Biomedical Engineering | National Electro-Medical Equipment Maintenance Workshop & Training Center and Technical College of Allied Health | Specialized government hospitals | 2026
- Technical Workshop on Biomaterials-Based Wet Lab | BUET, Oct 2025
- Data Analysis with Python (80 hrs) | CCC, KUET (EDGE Project, Bangladesh Computer Council | Nov–Dec 2024
- Workshop: Research Design & Thesis Writing using LaTeX | Advanced Bio-Engineering Club, KUET
- Workshop: Crafting Innovation-Entrepreneurship & Hardware in Fab Lab Ecosystem | KUET, Feb 2024

CERTIFICATIONS

- AI for Medical Diagnosis, DeepLearning.AI (Coursera) | Jun 2025
- Supervised Machine Learning: Regression and Classification | *DeepLearning.AI* | May 2025
- Secondary 2 Upper Intermediate (English) | *British Council* | 2018

TECHNICAL SKILLS

- **Laboratory & Characterization:** Hydrogel preparation; freeze-thaw crosslinking; freeze-drying; swelling; water retention; gel fraction; degradation; pH responsiveness; antimicrobial testing; drug loading/release; WVTR; UTM; FTIR; UV-Vis; TGA.
- **Computational & Simulation:** Python; MATLAB; COMSOL Multiphysics; OriginPro; Ansys.
- **Others:** Latex, MS Word, MS PowerPoint, MS Excel, Canva

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- Vice President, KUET Career Club | Sep 2025 to Sep 2026
- Senior Executive, IT & Resources, KUET Career Club | *Jun 2024 to Sep 2025* | IT & Designs
- Junior Executive, IT & Resources, KUET Career Club| *Mar 2023 to Jun 2024* | IT & Designs, Public Relations
- Junior Training Coordinator, Advanced Bio-Engineering Club of KUET | Apr 2023 to Sep 2026
- Campus Ambassador, Turtle Venture | Jan 2024 to Jan 2025
- Champion, Darts (Female) and Organizer, BME Indoor 4.0, Biomedical Engineering Association, KUET | 2025

HONORS & AWARDS

- Dean's Award, GPA 3.89/4.00 (3rd year), KUET, Sep 2026
- Dean's Award, GPA 3.83/4.00 (1st year), KUET, Sep 2023
- Best Performer (Crest), KUET Career Club, Senior and Junior Executive
- Talentpool Scholarship in Primary School Certificate Examination (PSC) by Board of Primary Education, Dhaka
- General Scholarship in Secondary School Certificate Examination (SSC) and Higher Secondary Certificate Examination (HSC) by Board of Secondary and Intermediate Education, Dhaka

REFERENCES

- Dr. Muhammad Muinul Islam, Professor, Department of Biomedical Engineering, Khulna University of Engineering & Technology (KUET) | Contact: *mmi@bme.kuet.ac.bd*
- Kazi Zannatul Ferdowshee, Assistant Professor, Department of Biomedical Engineering, Khulna University of Engineering & Technology (KUET) | Contact: *ferdwushee@bme.kuet.ac.bd*